DURABIN: MULTISCALE MODELING OF BIOLOGICAL MEMBRANES

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ABSTRACT

Many essential biological processes, such as protein folding and membrane remodeling, are inherently multiscale, going from atomic-level interactions to mesoscopic structures. Structural biology techniques and molecular dynamics simulations allow studying systems at the atomic resolution but the mesoscopic scale is more difficult to explore. In order to tackle this problem, multiscale computational methodologies have been developed, where data obtained from quantum mechanics (QM) calculations and all atom molecular dynamics (AA-MD) are used to parametrize and fine tune coarser models such as coarse-grained molecular dynamics (CG-MD) or Brownian dynamics (BD).

DURABIN is a framework developed over the past decade that integrates multiscale modeling with real-time visualization using the UNITY game engine. It enables users to construct, simulate, and explore systems with very little computational expertise. The simulations are done at the ultra-coarse-grain level and proteins are modeled as chains of rigid bodies connected by joints. The interactions between proteins and the membrane are modeled using the IMPALA force field while the interactions between proteins are modeled through colliders and a potential derived from all atom molecular dynamics simulations. The dimerization and diffusion of MMP14 proteins in a membrane are studied as a function of the parameters of interaction. Different biological conditions can be tested by changing the strength of interaction as well as the rate of activation and inactivation of the proteins.